Digital Media Fundamentals and Visual Communication

LIU08106 · BLISS III | RIU08106 · BRIM III

How media are encoded and arranged shapes what users see and understand.

Unit 2 learning outcomes

- Explain pixel dimensions, raster graphics and vector graphics.
- Use PPI appropriately and calculate image data size.
- Compare RGB, CMYK, grayscale and bit depth.
- Apply hierarchy, typography, spacing, colour and accessible layout.

Judge both technical suitability and communication effectiveness.

Why technical choices matter

- A sharp picture on one device may be unclear when stretched or printed.
- A visually attractive poster can still hide its most important information.
- Archival displays require context and faithful representation.
- Good choices begin with source, users, device and purpose.

Design quality is not only appearance; users must be able to act on the information.

What is a raster image?

- A raster image contains a grid of pixels, each with a colour or tonal value.
- Width × height describes its pixel dimensions.
- For example, 1200 × 800 = 960,000 pixel positions.
- Raster representation suits photographs and document scans.

Zooming a small raster far beyond its available pixels creates blocky or soft results.

Pixel grid illustration

- Think of each pixel as a tile in a mosaic.
- At normal size many tiles visually merge into a picture.
- Strong enlargement exposes individual tiles.
- Resampling can create interpolated pixels but cannot guarantee missing source detail.

Source image detail matters more than a new size label.

What is a vector graphic?

- Vector graphics describe lines, curves and shapes mathematically.
- At different output sizes the drawing can be rendered again sharply.
- Good uses: logos, interface symbols, simple maps, flowcharts.
- SVG is a common vector format; PDF may contain raster and vector content.

Choose vector for scalable shapes; raster for realistic photographic tone.

Raster versus vector: choose by task

- Library book-cover photograph: raster, because it contains photographic tone.
- Library navigation icon: vector, because it has simple scalable geometry.
- Archival scan of a letter: raster, to preserve captured page detail.
- Diagram of a records lifecycle: vector, for clean lines and labels.

Neither representation is universally superior; match it to the information.

Pixel dimensions and megapixels

- Dimensions describe how many pixels exist horizontally and vertically.
- A 3000 × 2000 photograph contains 6,000,000 pixels.
- 6,000,000 pixels ≈ 6 megapixels (MP).
- Megapixels alone do not guarantee sharpness: focus and capture quality matter.

A larger image may contain more sampled detail, but quality also matters.

PPI is not the same as DPI

- PPI maps image pixels to a physical inch for capture or print planning.
- DPI usually refers to output dots a printer deposits.
- Changing a PPI tag without resampling need not alter the pixel count.
- A web browser commonly sizes images by CSS/layout rather than print PPI metadata.

Always state whether you mean source pixels, intended print density or printer dots.

Worked print-size calculation

- Source image: 2400 × 1600 pixels.
- At 300 PPI: print width = 2400 ÷ 300 = 8 inches.
- Print height = 1600 ÷ 300 ≈ 5.33 inches.
- In centimetres: about 20.3 × 13.5 cm.

The calculation plans image density, not guaranteed print quality.

Aspect ratio: avoid stretching

- 1920 × 1080 has a 16:9 aspect ratio.
- 1200 × 800 has a 3:2 aspect ratio.
- Proportional resizing keeps shape consistent.
- Cropping removes content; stretching distorts content.

An archival derivative must be labelled if a crop omits part of the source.

Downsampling, upsampling and cropping

- Downsampling removes pixels for a smaller derivative.
- Upsampling estimates extra pixels through interpolation.
- Cropping changes the visible content, potentially removing context.
- For access delivery, protect the source/master and record derivative transformations.

Do not confuse a convenient access copy with the complete original.

The RGB colour model

- RGB combines red, green and blue light.
- Typical 8-bit-per-channel RGB values range 0–255.
- RGB(255, 0, 0) gives red; RGB(0, 0, 0) black.
- RGB is widely used for monitors, websites and screen presentation.

Colour values need a suitable colour space/profile for predictable appearance.

CMYK and grayscale

- CMYK stands for cyan, magenta, yellow and key (black).
- Printing inks subtract parts of reflected light; print results depend on process and paper.
- Grayscale stores tonal differences without full colour.
- Screen RGB values do not translate perfectly into printed inks.

Select colour mode according to output and the requirement for fidelity.

Colour space and profiles

- A colour model describes components; a colour space interprets them more precisely.
- sRGB is common for web delivery.
- Different screens and printers can show the same numbers differently.
- Colour-managed workflows use profiles to reduce inconsistencies.

For historical materials, keep capture information and document editing decisions.

Bit depth: more levels, not more pixels

- An 8-bit channel represents 2
- 8 = 256 integer levels.
- A 24-bit RGB pixel commonly consists of three 8-bit channels.
- That gives 256^3 = 16,777,216 possible channel combinations.
- Higher per-channel depth helps retain tonal editing flexibility in some workflows.

Bit depth and pixel dimensions describe different image properties.

Image size estimate: the formula

- Uncompressed pixel bytes ≈ width × height × bits per pixel ÷ 8.
- Assume no row padding, extra metadata or container overhead.
- One MiB = 1,048,576 bytes.
- Saved JPEG or PNG sizes differ because compression and file structure matter.

Use this as a planning estimate, never as a promised saved-file size.

Worked file-size calculation

- Example: 1920 × 1080 pixels, 24 bits/pixel.
- 1920 × 1080 = 2,073,600 pixels.
- 2,073,600 × 3 bytes = 6,220,800 bytes.
- 6,220,800 ÷ 1,048,576 ≈ 5.93 MiB.

A 1920 × 1080 image can be far smaller on disk when efficiently compressed.

Visual communication begins with audience

- Identify who will read the message and what action they must take.
- A library event poster should highlight title, time, place and action.
- An archive display should highlight item identity, date and source context.
- Decorative elements should not compete with critical facts.

The essential message must be easy to locate before styling decisions are made.

Hierarchy and visual grouping

- Hierarchy controls reading order: title, main message, details, source.
- Proximity groups elements that belong together.
- Alignment makes lines and edges visually consistent.
- White space separates groups and reduces cognitive load.

If everything is equally bold, nothing is clearly important.

Typography for clarity

- Use one readable body typeface and a consistent heading style.
- Match size and spacing to device and viewing distance.
- Avoid all-capital body paragraphs and excessive centred text.
- Prefer actual selectable text over images of text where possible.

Typography is a communication decision, not merely decoration.

Contrast and accessible colour

- WCAG 2.2 AA generally calls for 4.5:1 contrast for normal text.
- Qualifying large text generally needs at least 3:1.
- There are stated exceptions, and colour contrast is only one accessibility measure.
- Do not communicate urgent status by red/green alone: add labels or patterns.

Use readable text, explicit labels and clear non-colour cues.

Example: improve a crowded library poster

- Weak design: four fonts, grey 10-point text, five equal headings.
- Improvement: one primary heading, clear schedule and strong action.
- Use left alignment, consistent margins and adequate spacing.
- Check the result on a mobile phone, not only a large desktop.

Test whether a first-time visitor can find event date and venue quickly.

Example: improve an archival exhibit card

- Weak design: large cropped photograph with no date or source.
- Improvement: accurate title, creator/date, provenance and rights note.
- State when the displayed image is an edited or cropped derivative.
- Use sharp, appropriately scaled scans and retain protected masters.

Readability should never come at the expense of source context.

Responsive design and small screens

- A three-column desktop image gallery may become one column on mobile.
- Text should remain legible without sideways scrolling.
- Menus and controls need clear labels and touch/keyboard access.
- Evaluate real content, not only the empty template.

A responsive layout protects the task when the device changes.

Worked design decision matrix

- Photograph of a collection item: raster + accurate description.
- Scalable interface icon: vector + accessible name.
- Print poster: plan physical dimensions and appropriate reproduction quality.
- Public web exhibit: optimise derivatives and preserve originals separately.

Technical and ethical decisions form one production workflow.

Guided practical: compare two layouts

- Select an authorised image and define your audience.
- Record dimensions, aspect ratio and intended output.
- Sketch two layouts, then choose hierarchy, fonts and colour.
- Check legibility, contrast, source attribution and peer feedback.

Demonstrate a justified design improvement, not merely a prettier appearance.

BLISS pathway: digital library notice

- Use a clear heading for the service or event.
- Include accurate time/place or the next action near the heading.
- Use a simple icon or suitable photograph with attribution.
- Ask another student to find the essential detail in ten seconds.

Measure whether the notice helps library users complete a task.

BRIM pathway: archival record display

- Start with a valid item identifier, title and date/context.
- Use a faithful access scan; preserve a source/master separately.
- If the display uses a crop or enhanced derivative, label it.
- Verify restrictions, permissions and information accuracy.

A trustworthy exhibition card communicates provenance as well as appearance.

Unit 2 quick review

- What is the difference between PPI and DPI?
- What is the pixel payload size of 800 × 600 at 24 bits/pixel?
- Why is a vector graphic suitable for a logo but not a photographic scan?
- What four changes would improve a hard-to-read poster?

Explain concepts in your own words and justify decisions using professional examples.

Your application task

- Create a one-page information visual for BLISS or BRIM users.
- Submit the design with a short rationale explaining technical and layout decisions.
- Include an accessibility check and source/permission information.
- Use the course submission instructions for deadlines and delivery.

The website can record personal progress but does not submit coursework.